

Cooling real gases

27

In Chapter 26, we considered how to model the properties of real gases using various corrections to the ideal gas model. In this chapter, we will use these results to explore some of the deviations from ideal gas behaviour which can be observed in practice, in particular with changes in the behaviour of a Joule expansion. Then we will introduce the Joule–Kelvin throttling process (which has no effect on an ideal gas, but which can lead to cooling of a real gas) and discuss how real gases can be liquefied.

27.1 The Joule expansion

We have discussed the properties of non-ideal gases in some detail. In this section, we will see how the intermolecular interactions in such gases lead to departures from ideal-gas behaviour for the Joule expansion. Recall from Section 14.4 that a Joule expansion is an irreversible expansion of a gas into a vacuum which can be accomplished by opening a tap connecting the vessel containing gas and an evacuated vessel (see Fig. 27.1). The entire system is isolated from its surroundings and so no heat enters or leaves. No work is done, so the internal energy U is unchanged. We are interested in finding out whether the gas warms, cools or remains at constant temperature in this expansion.

To answer this, we define the **Joule coefficient** μ_J using

$$\mu_J = \left(\frac{\partial T}{\partial V} \right)_U, \quad (27.1)$$

where the constraint of constant U is relevant for the Joule expansion. This partial differential can be transformed using eqn 16.67 and the definition of C_V to give

$$\mu_J = - \left(\frac{\partial T}{\partial U} \right)_V \left(\frac{\partial U}{\partial V} \right)_T = - \frac{1}{C_V} \left(\frac{\partial U}{\partial V} \right)_T. \quad (27.2)$$

Now the first law, $dU = TdS - pdV$, implies that

$$\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial S}{\partial V} \right)_T - p, \quad (27.3)$$

and using a Maxwell relation (eqn 16.53) this becomes

$$\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial p}{\partial T} \right)_V - p, \quad (27.4)$$

27.1 The Joule expansion	297
27.2 Isothermal expansion	299
27.3 Joule–Kelvin expansion	300
27.4 Liquefaction of gases	302
Chapter summary	303
Exercises	304

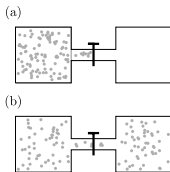


Fig. 27.1 The Joule expansion: (a) before opening the tap and (b) after opening the tap.

and hence

$$\mu_J = -\frac{1}{C_V} \left[T \left(\frac{\partial p}{\partial T} \right)_V - p \right]. \quad (27.5)$$

For an ideal gas, $p = RT/V$, $(\partial p/\partial T)_V = R/V$, and hence $\mu_J = 0$. Hence, as we found in Section 14.4, the temperature of an ideal gas is unchanged in a Joule expansion. For real gases, you always get cooling because of the attractive effect of interactions. This is because $C_V > 0$ and $(\partial U/\partial V)_T > 0$ and so $\mu_J = -\frac{1}{C_V}(\partial U/\partial V)_T < 0$. This can be understood physically as follows: when a gas freely expands into a vacuum, the time-averaged distance between neighbouring molecules increases and the magnitude of the potential energy resulting from the attractive intermolecular interactions is reduced. However, this potential energy is a negative quantity (because the interactions are attractive) and so the potential energy is actually *increased* (because it is made less negative).¹ Since U must be unchanged in a Joule expansion (no heat enters or leaves and no work is done), the kinetic energy must be reduced (by the same amount by which the potential energy rises) and hence the temperature falls.

¹Of course, at very high densities, the intermolecular interactions become repulsive rather than attractive, but at such a density one is probably dealing with a solid rather than a gas.

Example 27.1

Evaluate the Joule coefficient for a van der Waals gas.

Solution:

The equation of state is $p = RT/(V - b) - a/V^2$ and so

$$\left(\frac{\partial p}{\partial T} \right)_V = \frac{R}{V - b}, \quad (27.6)$$

and hence

$$\mu_J = -\frac{1}{C_V} \left[\frac{RT}{V - b} - \frac{RT}{V - b} + \frac{a}{V^2} \right] = -\frac{a}{C_V V^2}. \quad (27.7)$$

The temperature change in a Joule expansion from V_1 to V_2 can be evaluated simply by integrating the Joule coefficient as follows:

$$\Delta T = \int_{V_1}^{V_2} \mu_J dV = - \int_{V_1}^{V_2} \frac{1}{C_V} \left[T \left(\frac{\partial p}{\partial T} \right)_V - p \right] dV. \quad (27.8)$$

Example 27.2

Evaluate the change in temperature for a van der Waals gas which undergoes a Joule expansion from volume V_1 to volume V_2 .

Solution:

Using eqn 27.8, we have that

$$\Delta T = -\frac{a}{C_V} \int_{V_1}^{V_2} \frac{dV}{V^2} = -\frac{a}{C_V} \left(\frac{1}{V_1} - \frac{1}{V_2} \right) < 0 \quad (27.9)$$

since $V_2 > V_1$ in an expansion.

27.2 Isothermal expansion

Consider the isothermal expansion of a non-ideal gas. Equation 27.4 states that

$$\left(\frac{\partial U}{\partial V}\right)_T = T \left(\frac{\partial p}{\partial T}\right)_V - p, \quad (27.10)$$

so that the change of U in an isothermal expansion is

$$\Delta U = \int_{V_1}^{V_2} \left[T \left(\frac{\partial p}{\partial T}\right)_V - p \right] dV. \quad (27.11)$$

- For an ideal gas², $\Delta U = 0$.

- For a van der Waals gas, $\Delta U = \int_{V_1}^{V_2} \frac{a}{V^2} dV = a(1/V_1 - 1/V_2)$.

²For an ideal gas, $p = RT/V$, $(\partial p / \partial T)_V = R/V$, and hence $T \left(\frac{\partial p}{\partial T}\right)_V - p = 0$.

Note that U depends on a , not b (it is influenced by the intermolecular interactions but does not ‘care’ that they have non-zero size). Note also that for large volumes, U becomes independent of V and one recovers the ideal gas limit.

Example 27.3

Calculate the entropy of a van der Waals gas.

Solution:

The entropy S can be written as a function of T and V so that $S = S(T, V)$. Hence

$$\begin{aligned} dS &= \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV \\ &= \frac{C_V}{T} dT + \left(\frac{\partial p}{\partial T}\right)_V dV, \end{aligned} \quad (27.12)$$

where eqns 16.68 and 16.53 have been used to obtain the second line. For the van der Waals gas, we can write $(\partial p / \partial T)_V = R/(V - b)$, and hence we have that

$$S = C_V \ln T + R \ln(V - b) + \text{constant}. \quad (27.13)$$

Note that the entropy depends on the constant b , but not a . Entropy ‘cares’ about the volume occupied by the molecules in the gas (because this determines how much available space there is for the molecules to move around in, and this in turn determines the number of possible microstates of the system) but not about the intermolecular interactions.

27.3 Joule–Kelvin expansion

The Joule expansion is a useful conceptual process, but it is not much practical use for cooling gases. Gas slightly cools when it is expanded into a second evacuated vessel, but what do you do with it then? What is wanted is some kind of flow process where warm gas can be fed into some kind of a ‘cooling machine’ and cold gas (or better still, cold liquid) emerges from the other end. Such a process was discovered by James Joule and William Thomson (later Lord Kelvin) and is known as a **Joule–Thomson expansion** or a **Joule–Kelvin expansion**.

Consider a steady flow process in which gas at high pressure p_1 is forced through a throttle valve or a porous plug to a lower pressure p_2 . This is illustrated in Fig. 27.2. Consider a volume V_1 of gas on the high-pressure side. Its internal energy is U_1 . To push the gas through the constriction, the high pressure gas behind it has to do work on it equal to $p_1 V_1$ (since the pressure p_1 is maintained on the high-pressure side of the constriction). The gas expands as it passes through to the low-pressure region and now occupies volume V_2 which is larger than V_1 . It has to do work on the low-pressure gas in front of it which is at pressure p_2 and hence this work is $p_2 V_2$. The gas may change its temperature in the process and hence its new internal energy is U_2 . The change in internal energy ($U_2 - U_1$) must be equal to the work done on the gas ($p_1 V_1$) minus the work done by the gas ($p_2 V_2$). Thus

$$U_1 + p_1 V_1 = U_2 + p_2 V_2 \quad (27.14)$$

or equivalently

$$H_1 = H_2, \quad (27.15)$$

so that it is *enthalpy* that is conserved in this flow process.

Since we are now interested in how much the gas changes temperature when we reduce its pressure at constant enthalpy, we define the Joule–Kelvin coefficient by

$$\mu_{JK} = \left(\frac{\partial T}{\partial p} \right)_H. \quad (27.16)$$

This can be transformed using the reciprocity theorem (eqn C.42) and the definition of C_p to give

$$\mu_{JK} = - \left(\frac{\partial T}{\partial H} \right)_p \left(\frac{\partial H}{\partial p} \right)_T = - \frac{1}{C_p} \left(\frac{\partial H}{\partial p} \right)_T. \quad (27.17)$$

Now the relation $dH = TdS + Vdp$ implies that

$$\left(\frac{\partial H}{\partial p} \right)_T = T \left(\frac{\partial S}{\partial p} \right)_T + V, \quad (27.18)$$

and using a Maxwell relation (eqn 16.54) this becomes

$$\left(\frac{\partial H}{\partial p} \right)_T = -T \left(\frac{\partial V}{\partial T} \right)_p + V, \quad (27.19)$$

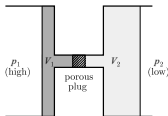


Fig. 27.2 A throttling process.

and hence

$$\mu_{JK} = \frac{1}{C_p} \left[T \left(\frac{\partial V}{\partial T} \right)_p - V \right] \quad (27.20)$$

The change in temperature for a gas following a Joule-Kelvin expansion from pressure p_1 to pressure p_2 is given by

$$\Delta T = \int_{p_1}^{p_2} \frac{1}{C_p} \left[T \left(\frac{\partial V}{\partial T} \right)_p - V \right] dp. \quad (27.21)$$

Since $dH = TdS + Vdp = 0$, the entropy change is

$$\Delta S = - \int_{p_1}^{p_2} \frac{V}{T} dp, \quad (27.22)$$

and for an ideal gas this is $R \ln(p_1/p_2) > 0$. Thus this is an irreversible process.

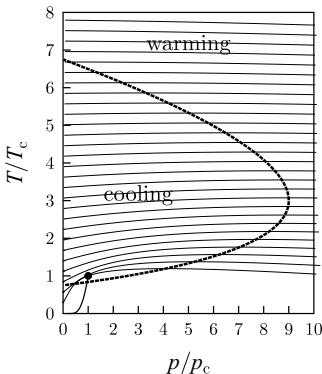


Fig. 27.3 The inversion curve of the van der Waals gas is shown as the heavy dashed line. The isenthalps (lines of constant enthalpy) are shown as thin solid lines. When the gradients of the isenthalps on this diagram are positive, then cooling can be obtained when pressure is reduced at constant enthalpy (i.e. in a Joule-Kelvin expansion). Also shown (as a solid line near the bottom left-hand corner of the graph which terminates at the dot) is the line of coexistence of liquid and gas (from Fig. 26.6) ending in the critical point ($p = p_c$, $T = T_c$, shown by the dot).

Whether the Joule-Kelvin expansion results in heating or cooling is more subtle and in fact μ_{JK} can take either sign. It is convenient to consider when μ_{JK} changes sign, and this will occur when $\mu_{JK} = 0$, i.e. when $T(\partial V/\partial T)_p - V = 0$, or equivalently

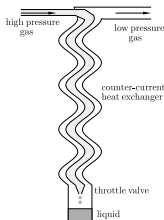
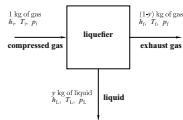
$$\left(\frac{\partial V}{\partial T} \right)_p = \frac{V}{T}. \quad (27.23)$$

^4He	H_2	N_2	Ar	CO_2
43	204	607	794	1275

Table 27.1 The maximum inversion temperature in Kelvin for several gases.

This equation defines the so-called **inversion curve** in the T - p plane. This is plotted for the van der Waals gas in Fig. 27.3 as a heavy solid line. The lines of constant enthalpy are also shown and their gradients change sign when they cross the inversion curve. When the gradient of the isenthalps on this diagram are positive, then cooling can be obtained when pressure is reduced at constant enthalpy (i.e. in a Joule-Kelvin expansion).

A crucial parameter is the maximum inversion temperature, below which the Joule-Kelvin expansion can result in cooling. These are listed for several real gases in Table 27.1. In the case of helium, this temperature is 43 K, so helium gas must be cooled to below this temperature by some other means before it can be liquefied using the Joule-Kelvin process.

**Fig. 27.4** A schematic diagram of a liquefier.**Fig. 27.5** A block diagram of the liquefaction process.

27.4 Liquefaction of gases

For achieving the liquefaction of gases, the Joule-Kelvin process is extremely useful, though it must be carried out below the maximum inversion temperature of the particular gas in question. A schematic diagram of a liquefier is shown in Fig. 27.4. High-pressure gas is forced through a throttle valve, resulting in cooling by the Joule-Kelvin process. Low-pressure gas plus liquid results, and the process is made more efficient by use of a counter-current heat exchanger by which the outgoing cold low-pressure gas is used to precool the incoming warm high pressure gas, helping to ensure that by the time it reaches the throttle valve the incoming high-pressure gas is already as cool as possible and at least at a temperature such that the Joule-Kelvin effect will result in cooling.

We can consider the liquefier as a ‘black box’ into which you put 1 kg of warm gas and get out y kg of liquid, as well as $(1-y)$ kg of exhaust gas (see Fig. 27.5). The variable y is the efficiency y of a liquefier, i.e. the mass fraction of incoming gas which is liquefied. Since enthalpy is conserved in a Joule-Kelvin process, we have that

$$h_i = y h_L + (1-y) h_f, \quad (27.24)$$

where h_i is the specific enthalpy of the incoming gas, h_L is the specific enthalpy of the liquid, and h_f is the specific enthalpy of the outgoing gas. Hence the efficiency y is given by

$$y = \frac{h_f - h_i}{h_f - h_L} \quad (27.25)$$

For an efficient heat exchanger, the temperature of the compressed gas T_i and the exhaust gas T_f will be the same. We also have that $p_f = 1 \text{ atm}$, and T_L is fixed (because the liquid will be in equilibrium with its vapour). Therefore h_f and h_L are fixed. The only parameter to vary is then h_i and to maximize y we must minimize h_i , i.e.

$$\left(\frac{\partial h_i}{\partial p_i}\right)_{T_i} = 0 \quad (27.26)$$

and since $(\partial h / \partial p)_T = -(1/C_p)\mu_{JK}$, we therefore require that

$$\mu_{JK} = 0. \quad (27.27)$$

This means that it is best to work the liquefier right on the inversion curve ($\mu_{JK} = 0$) for maximum efficiency.

Most gases could be liquefied by the end of the nineteenth century, but the modern form of gas liquefier dates back to the work of the German chemist Karl von Linde (1842–1934) who commercialized liquid-air production in 1895 using the Joule–Kelvin effect with a counter-current heat exchanger (as shown in Fig. 27.4; this is known as the **Linde process**) and discovered various uses of liquid nitrogen. James Dewar (1842–1923) was the first to liquefy hydrogen using the Linde process in 1898, and in 1899 got it to go solid. Dewar was also the first, in 1891, to study the magnetic properties of liquid oxygen. The Dutch physicist Heike Kamerlingh Onnes (1853–1926) was the first to produce liquid helium in 1908 by a similar process, precooling the helium gas using liquid hydrogen. Using liquid helium, he then discovered superconductivity in 1911, and was awarded the Nobel Prize in 1913 for ‘his investigations on the properties of matter at low temperatures which led, inter alia, to the production of liquid helium’.

Chapter summary

- The Joule expansion results in cooling for non-ideal gases because of the attractive interactions between molecules.
- The entropy of a gas depends on the non-zero size of molecules.
- The Joule–Kelvin expansion is a steady flow process in which enthalpy is conserved. It can result in either warming or cooling of a gas. It forms the basis of many gas liquefaction techniques.

Exercises

(27.1) (a) Derive the following general relations

$$(a) \quad \left(\frac{\partial T}{\partial V}\right)_U = -\frac{1}{C_V} \left[T \left(\frac{\partial p}{\partial T}\right)_V - p \right],$$

$$(b) \quad \left(\frac{\partial T}{\partial V}\right)_S = -\frac{1}{C_V} T \left(\frac{\partial p}{\partial T}\right)_V,$$

$$(c) \quad \left(\frac{\partial T}{\partial p}\right)_H = \frac{1}{C_p} \left[T \left(\frac{\partial V}{\partial T}\right)_p - V \right].$$

In each case the quantity on the left hand side is the appropriate thing to consider for a particular type of expansion. State what type of expansion each refers to.

(b) Using these relations, verify that for an ideal gas $(\partial T/\partial V)_U = 0$ and $(\partial T/\partial p)_H = 0$, and that $(\partial T/\partial V)_S$ leads to the familiar relation $pV^\gamma = \text{constant}$ along an isentrope.

(27.2) In a Joule-Kelvin liquefier, gas is cooled by expansion through an insulated throttle – a simple but inefficient process with no moving parts at low temperature. Explain why enthalpy is conserved in this process. Deduce that

$$\left(\frac{\partial T}{\partial p}\right)_H = \frac{1}{C_p} \left[T \left(\frac{\partial V}{\partial T}\right)_p - V \right].$$

Estimate the highest starting temperature at which the process will work for helium at low densities, on the following assumptions:

(i) the pressure is given at low densities by a virial expansion of the form

$$\frac{PV}{RT} = 1 + \left(b - \frac{a}{RT}\right) \left(\frac{1}{V}\right) + \dots,$$

and

(ii) the Boyle temperature a/bR (the temperature at which the second virial coefficient vanishes) is known from experiment to be 19 K for helium.

[Hint: One method of solving this problem is to remember that p is easily made the subject of the equation of state and one can then use $(\partial V/\partial T)_p = -(\partial p/\partial T)_V / (\partial p/\partial V)_T$.]

(27.3) For a gas obeying Dieterici's equation of state

$$p(V-b) = RTe^{-a/RTV},$$

for 1 mole, prove that the equation of the inversion curve is

$$p = \left(\frac{2a}{b^2} - \frac{RT}{b}\right) \exp\left(\frac{1}{2} - \frac{a}{RTb}\right),$$

and hence find the maximum inversion temperature $T_{\max}$.

(27.4) Show that the equation for the inversion curve of the Dieterici gas in reduced units is

$$\tilde{P} = (8 - \tilde{T}) \exp\left[\frac{5}{2} - \frac{4}{\tilde{T}}\right],$$

and sketch it in the $\tilde{T}$ - $\tilde{P}$ plane.

(27.5) Why is enthalpy conserved in steady flow processes? A helium liquefier in its final stage of liquefaction takes in compressed helium gas at 14 K, liquefies a fraction α , and rejects the rest at 14 K and atmospheric pressure. Use the values of enthalpy H of helium gas at 14 K as a function of pressure p in the table below to determine the input pressure which allows α to take its maximum value, and determine what this value is.

p (atm)	0	10	20	30	40
H (kJ kg ⁻¹)	87.4	78.5	73.1	71.8	72.6

[Enthalpy of liquid helium at atmospheric pressure = 10.1 kJ kg⁻¹].